Question

Dissolve $MnCl_2 \cdot 4H_2O$ in ethanol, and add an aqueous solution of $Htopca$ (a carboxylic acid derivative with the molecular formula $C_5H_6N_2O_3$). After mixing and stirring at 80 ℃ for 2 h, filter. The filtrate is slowly evaporated at room temperature, and a yellow blocky crystal precipitates from the solution. Thermogravimetric analysis of $\mathbf{X}$ under a nitrogen atmosphere shows that $\mathbf{X}$ loses a% of its weight between 65 and 120 ℃, corresponding to the loss of all crystal water and structural water; continue heating to 250 ℃, the weight loss is b%; continue heating to 850 ℃, the weight loss is further 14.04%, and the subsequent solid mass no longer changes. Chemical analysis of the residual solid shows that the solid contains two substances, and the mass fraction of Mn in the mixture is 46.17%. The mixed gas released during the second weight loss is analyzed, and the mixed gas contains three gas molecules, of which only propyne contains carbon. After cooling the mixed gas, it is fully absorbed by bromine water. The mass of the remaining gas is 37.94% of the original gas mass, and the remaining gas changes from colorless to reddish-brown in the air. It is known that $\mathbf{X}$ is a mononuclear Mn complex, and the oxidation state of Mn does not change during the thermogravimetric analysis.

Given a+b=58.07, which of the following statements are correct:

1. The sum of the number of all atoms in the chemical formula of $\mathbf{X}$ is less than 45

2. The three gases produced by the second weight loss are in a ratio of 1:1:2

3. In the total reaction equation for the second and third steps of weight loss, the sum of the coefficients on the right side is less than 12

4. The final product of the weight loss can react with concentrated sulfuric acid, producing only one gas

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: A

Detailed Explanation

Since the remaining gas changes from colorless to reddish-brown in air, and considering that Htopca contains nitrogen, it is speculated that the remaining gas contains NO; since the gas first underwent cooling, it is speculated that it may also contain $\mathrm{H_2O}$ before cooling.

Therefore, the composition of the mixed gas is likely propyne + NO + $\mathrm{H_2O}$, or propyne + NO + another gas.

CHECKPOINT 1 PTS

The composition of the mixed gas is likely propyne + NO + $\mathrm{H_2O}$, or propyne + NO + another gas.

Since the mass of the remaining gas is 37.94% of the original gas, assuming it is all NO, then:

$$\mathrm{Mr(remaining\ 1)} = (14.01 + 16.00)/0.3794 - (14.01 + 16.00) = 49.09$$

Subtracting the relative molecular mass of one molecule of propyne yields:

$$\mathrm{Mr(remaining\ 2)} = 49.09 - 40.06 = 9.03$$

This corresponds exactly to 0.5 $\mathrm{H_2O}$, which is consistent with the prediction.

Therefore, the gas lost in the second weight loss is $\mathrm{2C_3H_4 + 2NO + H_2O}$, so statement 2 is incorrect.

CHECKPOINT 2 PTS

The ratio of the gas lost in the second weight loss is $C_3H_4 : NO : H_2O = 2 : 2 : 1$

Since Mn does not change during the process, the Mn in the residual solid should be an oxide of the corresponding valence state, namely MnO. Combined with the mass fraction, we have $Mr(\text{remaining}) = 54.94/0.4617 - 54.94 - 16 = 48.06$, which is found to be exactly the relative mass of 4 carbons, so the molecular composition of the residual solid is $MnO + 4C$. The reaction of elemental carbon with concentrated sulfuric acid can produce CO_2 and SO_2, so statement 4 is incorrect.

CHECKPOINT 1 PTS

The residual solid is $MnO : C = 1 : 4$

Therefore, the molecular weight of $\mathbf{X}$ is $Mr = 119.00/(1 - 0.5807 - 0.1404) = 426.66$

The first and second weight losses $M = 426.66 \times 0.5807 = 247.76$. Removing $2C_3H_4 + 2NO + H_2O$ leaves approximately 90, so the first loss is $5H_2O$

The third weight loss $M = 426.66 \times 0.1404 = 59.90$, corresponding to $2NO$

CHECKPOINT 1 PTS

The third loss is NO

Therefore, the molecular formula of $\mathbf{X}$ is $MnC_{10}H_{10}N_4O_6 \cdot 5H_2O$, i.e., $Mn(topca)_2(H_2O)_5$

CHECKPOINT 2 PTS

The molecular formula of $\mathbf{X}$ is $MnC_{10}H_{10}N_4O_6 \cdot 5H_2O$, i.e., $Mn(topca)_2(H_2O)_5$

The sum of the number of all atoms in the chemical formula of $\mathbf{X}$ is 46, so statement 1 is incorrect.

The second and third steps are the loss of $2C_3H_4 + 4NO + H_2O$ from 1 $\mathbf{X}$, producing $MnO + 4C$. The equation is $Mn(topca)_2 \rightarrow 2C_3H_4 + 4NO + H_2O + MnO + 4C$, and the sum of the coefficients on the right side is 12, so statement 3 is incorrect.

CHECKPOINT 1 PTS

The total equation for the second and third weight loss steps is $Mn(topca)_2 \rightarrow 2C_3H_4 + 4NO + H_2O + MnO + 4C$

Choose A